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Effects of Exercise on Hyperactivity/Impulsivity and Inhibitory Control at Behavioral and Electrophysiological Levels in ADHD: A Systematic Review and Meta-Analysis

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Abstract

Objective: This study aimed to assess the impact of exercise on hyperactivity/impulsivity, inhibitory control, and inhibition-related event-related potential (ERP) components in individuals with ADHD.

Method: A systematic search identified relevant studies, and methodological quality was assessed using the Revised Cochrane Risk-of-Bias tool for randomized trials (RoB 2) and the Risk of Bias in Non-Randomized Studies of Interventions (ROBINS-I), with data analysis conducted using Stata software.

Results: A total of 36 studies (38 comparisons) were included, comprising 10 acute and 26 chronic exercise interventions. Exercise yielded a small-to-moderate improvement in inhibitory control but showed no significant effects on hyperactivity/impulsivity or inhibition-related N2 and P3 components. Subgroup analyses of inhibitory control revealed significant moderating effects of age (children/adolescents), intervention type (chronic interventions), frequency (three sessions per week), control condition (sedentary or no-intervention groups), and study quality (studies with moderate or high risk of bias).

Conclusion: Exercise enhances inhibitory control in individuals with ADHD, with the effect being especially pronounced in children and adolescents. Chronic interventions and a frequency of three sessions per week appear to be most beneficial. However, it shows no significant effect on hyperactivity/impulsivity or inhibition-related N2 and P3 components. The impact of exercising should not be overestimated. (*J. of Att. Dis. 2026; 30(5) 677-693*)

Keywords

attention deficit hyperactivity disorder, exercise, inhibitory control, behavioral performance

Introduction

Attention deficit hyperactivity disorder (ADHD) is a common neurodevelopmental disorder marked by inattention, hyperactivity, and impulsivity, typically emerging in childhood and often persisting into adulthood (American Psychiatric Association, 2022). Its global prevalence is about 8% in children/adolescents (Ayano et al., 2023) and 2.58% among adults (P. Song et al., 2021). ADHD negatively impacts academic performance (Jangmo et al., 2019), social functioning (Nijmeijer et al., 2008), and emotional regulation (Stern et al., 2020). Additionally, research has linked ADHD to an increased risk of suicidal behavior (K.-L. Huang et al., 2018) and substance use disorders (Lee et al., 2011).

Inhibitory control deficits are a core cognitive mechanism underlying impulsivity and executive dysfunction in ADHD (Barkley, 1997). It refers to regulating attention, behavior, thoughts, and emotions to suppress internal tendencies or

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external distractions, with two subtypes: response inhibition (suppressing automatic motor responses) and interference control (avoiding distractions from irrelevant stimuli) (Diamond, 2013). Impaired inhibitory control is closely linked to hyperactive and impulsive behaviors, such as restlessness, excessive talking, and frequent interruptions (Johnstone & Clarke, 2009). Behavioral studies show that individuals with ADHD have higher error rates in tasks like Go/No-Go (Uebel et al., 2010), Flanker (Marquardt et al., 2018), and Continuous Performance Test (Doehnert et al., 2010), indicating difficulty suppressing impulsive responses and filtering out irrelevant information. At the electroencephalographic (EEG) level, the event-related potentials (ERPs) closely associated with inhibitory control are N2 and P3. The N2 component refers to a negative deflection in the frontocentral region occurring within the 250 to 400 ms time window after stimulus presentation under No-Go or incongruent conditions, with its primary neural source located in the anterior cingulate cortex (Gang & Tie-jun, 2010). It reflects the process of conflict monitoring (Groom & Cragg, 2015; Randall & Smith, 2011). The P3 component related to inhibition appears as a positive deflection in the central region within the 300 to 600 ms time window after stimulus presentation, with its primary neural sources located in the inferior frontal gyrus, precentral gyrus, and supplementary motor area (J. L. Smith et al., 2013). It is considered a marker of both cognitive and response inhibition (J. L. Smith et al., 2008). Neurophysiological evidence supports these findings, with reduced N2 and P3 amplitudes in No-Go trials among individuals with ADHD, suggesting impaired conflict monitoring (Broyd et al., 2005; Johnstone et al., 2009) and inhibitory processing (Chmielewski et al., 2018; Zarka et al., 2021). A neuroimaging meta-analysis revealed reduced volume and function in the ventrolateral prefrontal cortex/insular-striatal regions in ADHD (Norman et al., 2016), which may contribute to the core neurocognitive deficits commonly observed in children with the disorder.

Exercise has gained attention as an ADHD intervention due to its cost-effectiveness, accessibility, and non-invasive nature. Regarding hyperactivity/impulsivity, some studies suggest yoga alleviates ADHD symptoms in children (Luo et al., 2023), while others do not confirm this effect (Benzing & Schmidt, 2019; Bustamante et al., 2016; F. Sun et al., 2024), highlighting inconsistencies. Inhibitory control improvements have been observed in children with ADHD following aquatic exercise (Chang et al., 2014), ball games, and aerobic exercise (Memarmoghaddam et al., 2016), with further benefits from aerobic exercise combined with cognitive tasks (Liang et al., 2022), exergaming (Benzing et al., 2018), and table tennis (Pan et al., 2016). These effects may be mediated by brain changes, increased serotonin, dopamine, brain-derived neurotrophic factor (BDNF), and enhanced neuroplasticity (Berwid & Halperin, 2012; Chan et al., 2022). However, some studies report no significant effects on inhibitory control in children with ADHD (Geladé et al., 2017; Ludyga et al., 2023; F. Sun et al., 2024). Similarly, findings in adults with ADHD have also been mixed, with some studies failing to demonstrate consistent benefits (K. Fritz & O'Connor, 2022; Kouhbanani et al., 2023). These inconsistencies call for systematic reviews and meta-analyses to better evaluate exercise interventions and identify moderating factors.

Although previous meta-analyses have examined the effects of exercise on ADHD symptoms and executive function in children (H. Huang et al., 2022; Liang et al., 2021; Seiffer et al., 2022; W. Sun et al., 2022), several limitations remain. These studies often treat executive function as a unitary construct, with moderator analyses focusing on overall executive function and not isolating specific subdomains like inhibitory control. Inhibitory control, a core component of executive function linked to impulsivity (Bari & Robbins, 2013), has been insufficiently addressed. Additionally, most studies focus on children, neglecting adults with ADHD. Most research also relies on behavioral data, with limited electrophysiological evidence, restricting understanding of the neural mechanisms of exercise interventions. Finally, the dose-response relationship between exercise and inhibitory control is unclear, hindering optimal intervention development. This study aims to assess the effects of exercise on hyperactivity/impulsivity symptoms and inhibitory control at behavioral and electrophysiological levels in ADHD, explore the dose-response relationship, and identify potential moderators, providing a foundation for optimizing exercise-based interventions.

Methods

This study adhered to the PRISMA guidelines (Liberati et al., 2009) and was registered with PROSPERO (CRD42025637758).

Eligibility Criteria

Inclusion criteria were: (a) ADHD diagnosis based on the Diagnostic and Statistical Manual of Mental Disorders (DSM) or the International Classification of Diseases (ICD), with no age restrictions; (b) interventions involving exercise or physical activity, with no restrictions on type, either alone or in combination with usual treatment; (c) control conditions included medication, neurofeedback therapy, psychoeducation, sedentary activities, waitlist control, and regular physical activity; (d) outcomes including hyperactivity/impulsivity symptom questionnaires, behavioral tasks on inhibitory control, and corresponding event-related potentials (ERP); and (e) study design including randomized controlled trials (RCTs), non-randomized controlled trials (Non-RCTs), and crossover designs (CRD). Exclusion criteria were: (a) cross-sectional studies, (b) non-peer-reviewed articles (e.g.,

theses), (c) studies in languages other than Chinese or English, (d) conference abstracts, and (e) studies with unextractable or unconvertible outcome data.

Literature Search

This study conducted a literature search in PubMed, Web of Science, EBSCOhost, Embase, Cochrane Library, PsycINFO, and China National Knowledge Infrastructure (CNKI), covering all records from each database's inception to January 12, 2025. We used a combination of keywords and Medical Subject Headings (MeSH) terms, including "Attention-Deficit/Hyperactivity Disorder," "Exercise," "Physical Activity," "Inhibitory Control," "Response Inhibition," and "Interference Control." Search strategies were tailored to each database's requirements. In addition, the reference lists of all included studies and relevant reviews were manually screened by two researchers to identify additional eligible studies. Any disagreements regarding the inclusion of additional studies were resolved through discussion with the first author. The full search strategy, including Boolean operators and database-specific adaptations, is provided in Supplemental Material 1.

Screening and Data Extraction

The first author imported the relevant literature into EndNote 20 and created a reference library. Duplicate articles were removed, and two researchers independently conducted an initial screening by reviewing titles and abstracts. Full texts of the articles that passed the initial screening were then downloaded, and the final inclusion of studies was determined, along with the extraction of relevant data. Data extracted included study characteristics (i.e., author, publication time, country, and study design), participant details (i.e., age, sample size, diagnostic criteria, ADHD type, and medication usage), interventions (i.e., program, exercise session, frequency, duration, and exercise intensity), outcome measures (i.e., types of questionnaires, behavioral tasks, and ERP components, and key quality assessment elements. Literature screening and data extraction were independently conducted by two researchers. Any discrepancies were resolved through discussion with the first author.

Risk of Bias Assessment

The risk of bias in included studies was assessed using the Revised Cochrane Risk-of-Bias tool for randomized trials (RoB 2), the RoB 2 tool for crossover trials, and the Risk of Bias in Non-Randomized Studies of Interventions (ROBINS-I). Two independent researchers conducted the assessments, with disagreements resolved by discussion or consultation with the first author. For randomized controlled trials, RoB 2 was used to evaluate bias across five domains: randomization, deviations from intended interventions, missing outcome data, outcome measurement, and selection of reported results, with studies classified as having low risk, some concerns, or high risk (Sterne et al., 2019). For crossover trials, the adapted RoB 2 tool incorporated additional considerations such as carryover effects (The Cochrane Collaboration, 2021). For non-randomized studies, ROBINS-I evaluated bias across seven domains: confounding, participant selection, classification of interventions, deviations from intended interventions, missing data, outcome measurement, and selection of reported results, with studies categorized as low, moderate, serious, or critical risk (Sterne et al., 2016).

Statistical Analysis

The meta-analysis was performed using Stata SE 14.0 with the following treatments: (1) Inhibitory control at the behavioral level was assessed using key measures like No-Go error rates in the Go/No-Go task, converting accuracy to error rate when necessary; (2) In ERPs, inhibitory control is assessed through inhibition-related components, including the amplitudes of N2 and P3 in the fronto-central region under No-Go or incongruent conditions; (3) For RCTs with three study arms, the sample size of the control group was split equally (1:1) following Cochrane recommendations to minimize unit-of-analysis bias; (4) If symptom questionnaires were completed by both parents and teachers, all outcomes were included; (5) If both response inhibition and interference control were assessed, all relevant outcomes were included. The pre-to-post change in mean and standard deviation (SD) was calculated using the formulas: $\text{mean}_{\text{change}} = \text{mean}_{\text{post}} - \text{mean}_{\text{pre}}$; $\text{SD}_{\text{change}} = \text{square root } [\text{SDpre}^2 + \text{SDpos}^2 - (2r \times \text{SDpre} \times \text{SDpos})]$, where the correlation coefficient (r) was fixed at 0.5 (H. Huang et al., 2022).

A random-effects model was used for the pooled analysis. Heterogeneity was assessed using the Q statistic and I^2 statistic. The standardized mean difference (SMD) was estimated using Hedges' g, with effect sizes interpreted as large (≥ 0.8), moderate (0.5), and small (≤ 0.2), as defined by Hedges (1981). Subgroup analyses were conducted based on the following moderators: subject age (children/adolescents, adults), domains of inhibitory control (response inhibition, interference control), study design (RCT, CRD, non-RCT), exercise intervention type (acute: a one-time, short-duration exercise session; chronic: repeated and regular exercise training over an extended period of time), exercise duration (<12 weeks, ≥ 12 weeks), exercise frequency (2 times/week, 3 times/week, 5 times/week), exercise session length (≤ 50 min, >50 min), exercise intensity (moderate, moderate-to-vigorous, vigorous), motor skill type (closed-skill, open-skill), control group type (sedentary activity/no treatment, traditional aerobic exercise,

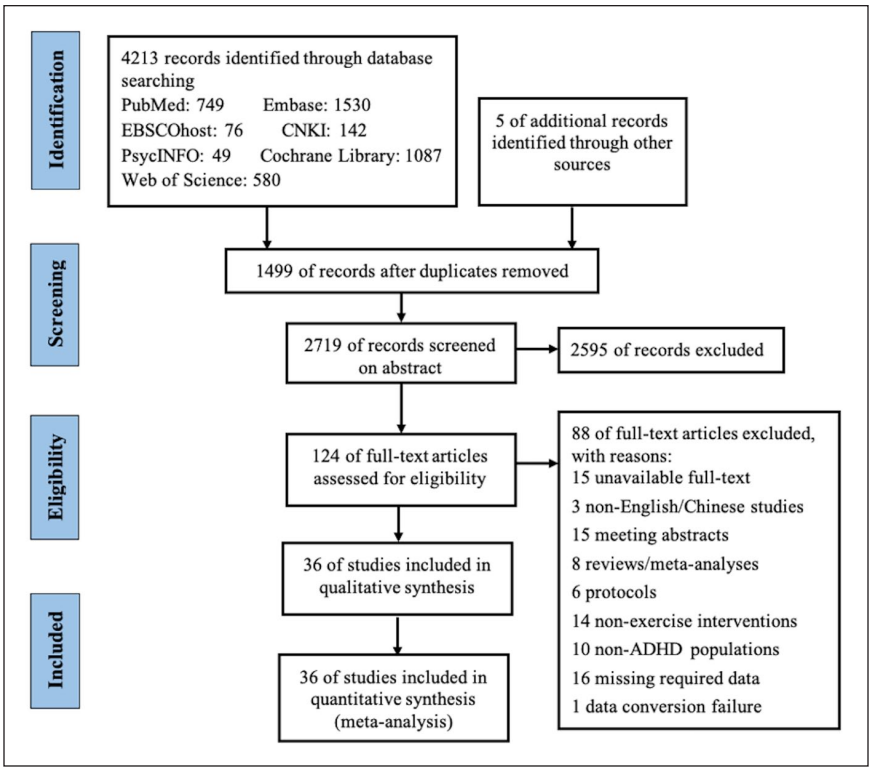


Figure 1. Flowchart of study identification, screening, and inclusion.

neurofeedback training), and study quality (low, moderate, high risk). Sensitivity analysis was conducted by sequentially excluding individual studies to assess the robustness of the meta-analysis results. Funnel plots and Egger's test were used to examine publication bias, with Egger's test applied only when more than 10 studies were available. When publication bias was detected, trim-and-fill methodology was applied for correction.

Results

Screening Results

The study selection process is shown in Figure 1. A total of 4,218 records were identified through electronic databases and manual searches. After removing 1,499 duplicates, 2,595 records were excluded based on title and abstract screening, leaving 124 studies. Of these, 88 were excluded for reasons such as unavailability of full text, non-Chinese/English publications, conference abstracts, reviews, non-exercise interventions, non-ADHD populations, missing data, or inability to convert data. Finally, 36 studies were included in the meta-analysis (Barudin-Carreiro et al., 2024; Benzing et al., 2018; Benzing & Schmidt, 2019; Bigelow et al., 2021; Bustamante et al., 2016; Chang et al., 2012, 2014; Chen et al., 2022; Converse et al., 2020; Dinu et al., 2023; K. M. Fritz & O'Connor, 2016; K. Fritz & O'Connor, 2022; Geladé et al., 2017; Ghadamgahi Sani et al., 2022; Hattabi et al., 2019; Jensen & Kenny, 2004; Ji et al., 2023; Kadri et al., 2019; Kouhbanani et al., 2023; Kuo, Nitsche, et al., 2024; Kuo, Sun, et al., 2024; Li et al., 2024; Liang et al., 2022; Ludyga et al., 2017, 2023; Luo et al., 2023; Memarmoghaddam et al., 2016; Nejati, 2021; Nejati & Derakhshan, 2021; Pan et al., 2016, 2019; S. D. Smith et al., 2019, 2020; Y. Song et al., 2022; F. Sun et al., 2024; Walters et al., 2024).

Study Characteristic

Table 1 summarizes the characteristics of the included studies. Of the 36 studies, 27 were RCTs (including two three-arm RCTs), 6 were crossover studies, and 3 were non-RCTs. Regarding age groups, 6 studies focused on adults (Converse et al., 2020; Dinu et al., 2023; K. M. Fritz & O'Connor, 2016; K. Fritz & O'Connor, 2022; Kouhbanani et al., 2023; Kuo, Nitsche, et al., 2024), while the remaining studies examined children or adolescents. In terms of language, 3 studies were published in Chinese (Chen et al., 2022; Li et al., 2024; Y. Song et al., 2022), while the rest were in English. Regarding exercise intervention types, 10 studies investigated acute exercise interventions (Barudin-Carreiro et al., 2024; Benzing et al., 2018; Bigelow et al., 2021; Chang et al., 2012; Dinu et al., 2023; K. M. Fritz & O'Connor, 2016; Kuo, Nitsche, et al., 2024; Kuo, Sun,

Table 1. Characteristics of Included Studies.

Study			Participants (IG vs. CG)					Intervention				
Author and year	Country	Design	Sample size (*n*)	Mean age or age range	Diagnostic methods	ADHD type (I/HI/C, unknown)	Medication user (%)	Program (IG vs. CG)	Type	Session (min/), frequency (times/week), duration (weeks)	Exercise intensity	Outcome measures
Barudin-Carreiro et al. (2024)	USA	RCT	7 vs. 8	9.69 ± 1.75 vs. 9.65 ± 1.19	NR	2/2/2,1 vs. 2/0/6	71.42 vs. 75	Walking vs. sitting	Acute	20	NR	①
Benzing et al. (2018)	Switzerland	RCT	24 vs. 22	10.46 ± 1.35 vs. 10.50 ± 1.41	ICD-10	NR	79.2 vs. 77.3	Exergaming vs. watching a movie	Acute	15	55%–90%HRmax	②
Benzing and Schmidt (2019)	Switzerland	RCT	28 vs. 23	10.46 ± 1.30 vs. 10.39 ± 1.44	ICD-10	NR	71.4 % vs. 73.9	Exergaming vs. non-intervention	Chronic	30/3/8	NR	③⑥
Bigelow et al. (2021)	Canada	CRD	16	11.38 ± 1.50	DSM-V	3/1/3,9	56.25	Cycling vs. reading	Acute	10	65%–85% HRmax	①
Bustamante et al. (2016)	USA	RCT	19 vs. 16	9.4 ± 2.2 vs. 8.7 ± 2.0	DSM-IV	NR	NR	Structured play vs. sedentary activity	Chronic	90/5/10	75% HRmax	④⑨
Chang et al. (2012)	Taiwan	RCT	20 vs. 20	10.45 ± 0.95 vs. 10.42 ± 0.87	DSM-IV	10/2/8 vs. 4/3/13	50 vs. 50	Running vs. watching a video	Acute	30	50%–70%HRR	①
Chang et al. (2014)	Taiwan	Non-RCT	14 vs. 13	8.19 ± 7.65 vs. 8.78 ± 8.33	DSM-IV	4/2/8 vs. 3/0/10	46.15 vs. 50	Aquatic exercise vs. non-intervention	Chronic	90/2/8	Moderate intensity	⑤
Chen et al. (2022)	China (in China)	RCT	32 vs. 32	8.37 ± 1.68 vs. 7.89 ± 2.13	DSM-V	5/10/17 vs. 6/8/18	NR	Cycling vs. sitting	Chronic	25/3/12	60%–80%HRmax	①⑯
Converse et al. (2020)	USA	RCT	9 vs. 5 vs. 7	20.70 ± 1.50	NR	12/1/7	76.19	Taichi vs. kickboxing vs. non-intervention	Chronic	60/2/7	NR	②⑪
Dinu et al. (2023)	UK	RCT	40 vs. 42	26.62 ± 5.26	NR	NR	NR	Yoga vs. cycling	Acute	10	/	⑦
K. M. Fritz and O'Connor (2016)	USA	CRD	32	18–33	NR	NR	NR	Cycling vs. resting	Acute	20	65% VO2peak	⑥
K. Fritz and O'Connor (2022)	USA	RCT	16 vs. 16	20.16 ± 1.46	DSM-V	NR	NR	Yoga vs. non-intervention	Chronic	90/2/6	NR	②
Geladé et al. (2017)	Netherlands	RCT	37 vs. 39	9.80 ± 1.96 vs. 9.96 ± 1.88	DSM-IV	NR	No	HIIT vs. NFB	Chronic	45/3/12	70%–80%HRmax	④
Ghadamgahi Sani et al. (2022)	Iran	RCT	20 vs. 20	7.50 ± 1.34 vs. 8.41 ± 0.85	DSM-V	NR	NR	Perceptual-motor exercise vs. NFB	Chronic	45/3/7	NR	⑥
Hattabi et al. (2019)	Tunisia	RCT	20 vs. 20	9.95 ± 1.31 vs. 9.75 ± 1.33	DSM-IV	4/6/10 vs. 5/4/11	NR	Swimming vs. non-intervention	Chronic	90/3/12	Moderate intensity	①
Jensen and Kenny (2004)	Australia	RCT	11 vs. 8	10.63 ± 1.78 vs. 9.35 ± 1.70	DSM-IV	2/0/9 vs. 1/0/7	9.09 vs. 12.5	Yoga vs. cooperative activity	Chronic	60/20	NR	⑫⑬
Ji et al. (2023)	Korea	RCT	16 vs. 14	9.00 ± 1.46 vs. 8.85 ± 1.63	NR	NR	NR	Exergaming vs. cycling	Chronic	50/3/12	60%–80% HRR	⑤⑯
Kadri et al. (2019)	Tunisia	RCT	20 vs. 20	14.5 ± 3.5 vs. 14.2 ± 3	NR	NR	NR	Taekwondo vs. non-intervention	Chronic	50/2/52	NR	①
Kouhbanani et al. (2023)	Iran	RCT	25 vs. 27	35.24 ± 11.49 vs. 35.40 ± 11.01	DSM-V	16/0/9 vs. 18/0/9	NR	Pilates training vs. non-intervention	Chronic	45/3/24	NR	⑥
Kuo, Nitsche, et al. (2024)	Taiwan	CRD	26	23.5 ± 3.2	DSM-IV	NR	NR	Cycling vs. watching a video	Acute	30	50%–70%HRR	④
Kuo, Sun, et al. (2024)	Taiwan	CRD	24	17.2 ± 2.2	DSM-V	1/2/21	NR	Cycling vs. watching a video	Acute	30	50%–70%HRR	④
Li et al. (2024)	China (in China)	Non-RCT	16 vs. 12	7.93 ± 1.61	DSM-V	NR	NR	Ball activity vs. non-intervention	Chronic	60/2/8	64%–95%HRmax	①
Liang et al. (2022)	China	RCT	40 vs. 40	8.37 ± 1.42 vs. 8.29 ± 1.27	DSM-V	22/7/11 vs. 19/6/15	No	Combined aerobic and neurocognitive exercise vs. non-intervention	Chronic	60/3/12	60%–80%HRmax	②
Ludyga et al. (2017)	Switzerland	CRD	16	12.8 ± 1.8	DSM-IV	NR	NR	Cycling vs. coordinative exercise vs. watching a video	Acute	20	65%–70%HRmax	②⑰
Ludyga et al. (2023)	Switzerland	RCT	29 vs. 27	10.0 ± 1.2 vs. 10.8 ± 1.2	DSM-V	NR	NR	Judo vs. non-intervention	Chronic	60/2/12	NR	⑤⑯⑰
Luo et al. (2023)	China	RCT	15 vs. 15	4.94 ± 0.70 vs. 5.07 ± 0.88	NR	NR	NR	Yoga vs. non-intervention	Chronic	10/2/16	NR	⑭
Memarmoghaddam et al. (2016)	Iran	RCT	19 vs. 17	8.31 ± 1.29 vs. 8.29 ± 1.31	NR	6/5/8 vs. 5/3/9	NR	Ball activity vs. non-intervention	Chronic	90/3/8	65%–80%HRR	①⑤

(continued)

Table 1. (continued)

Study			Participants (IG vs. CG)					Intervention				
Author and year	Country	Design	Sample size (*n*)	Mean age or age range	Diagnostic methods	ADHD type (I/HI/C, unknown)	Medication user (%)	Program (IG vs. CG)	Type	Session (min/), frequency (times/week), duration (weeks)	Exercise intensity	Outcome measures
Nejati (2021)	Iran	RCT	15 vs. 14	9.73 ± 1.94 vs. 9.21 ± 1.25	NR	NR	NR	Combined balanced and cognitive training vs. running	Chronic	40-50/3/4-5	NR	⑤
Nejati and Derakhshan (2021)	Iran	RCT	15 vs. 15	9.43 ± 1.43	DSM-V	5/2/8 vs. 6/1/8	NR	Exercise combining cognitive and virtual technology vs. running	Chronic	40-50/3/4-5	NR	⑤
Pan et al. (2016)	Taiwan	RCT	16 vs. 16	8.93 ± 1.49 vs. 8.87 ± 1.56	DSM-IV	NR	56.25 vs. 56.25	Table tennis vs. non-intervention	Chronic	70/2/12	NR	①
Pan et al. (2019)	Taiwan	Non-RCT	15 vs. 15	9.08 ± 1.43 vs. 8.90 ± 1.66	DSM-IV	NR	60 vs. 60	Table tennis vs. non-intervention	Chronic	70/2/12	NR	①
S. D. Smith et al. (2019)	USA	RCT	13 vs. 16	7.23 ± 1.42 vs. 7.06 ± 1.06	DSM-IV	5/2/4, 2 vs. 5/2/7, 2	15.4 vs. 18.8	Integrated brain, body, and social intervention vs. standard care	Chronic	120/3/15	NR	⑤⑯⑰
S. D. Smith et al. (2020)	USA	RCT	42 vs. 38	7.6 ± 0.9 vs. 7.2 ± 1.2	DSM-IV	12/7/15, 8 vs. 8/5/16, 9	19.05 vs. 18.42	Integrated brain, body, and social intervention vs. standard care	Chronic	90/3/15	NR	②
Song et al. (2022)	China (in China)	RCT	8 vs. 8	7.68 ± 0.56 vs. 7.53 ± 0.79	DSM-IV	6/0/2 vs. 5/1/2	NR	Football vs. non-intervention	Chronic	30/5/6	NR	①
Sun et al. (2024)	China	RCT	16 vs. 15 vs. 18	10.10 ± 1.83	NR	NR	NR	Game HIIT vs. Structured aerobic exercise vs. non-intervention	Chronic	30/2/8 for HIIT; 60/2/8 for aerobic exercise	Moderate-to-vigorous intensity	①⑮
Walters et al. (2024)	UK	CRD	34	9.9 ± 1.0	NR	NR	58.82	Coordinative exercise and game vs. resting	Acute	30	NR	①

Note. NR = not reported; IG = intervention group; CG = control group; RCT = Randomized Controlled Trial; CRD = Cross-over Randomized Design; Non-RCT = Non-Randomized Controlled Trial. ① Stroop Task (Stroop Color and Word Test), ② Flanker Task (Eriksen Flanker Task), ③ Simon Task (Simon Effect Task), ④ Stop-Signal Task, ⑤ Go/No-Go Task, ⑥ Continuous Performance Test (CPT), ⑦ Test of Variables of Attention (TOVA), ⑧ The German version of the Conners-3 scales, ⑨ The Disruptive Behavior Disorders Rating Scale, ⑩ The Conners' Parent Symptom Questionnaire, ⑪ The Conners' Adult ADHD Rating Scales – Self-Report: Long Version, ⑫ The Conners' Teacher Rating Scale–Revised: Long (CTRSR: L), ⑬ The Conners' Parent Rating Scale–Revised: Long (CPRS–R:L), ⑭ SNAP-IV Rating Scale, ⑮ The Attention-Deficit/Hyperactivity-symptoms and Normal behaviors (SWAN) rating scale, ⑯ ERP (N2), ⑰ ERP (P3).

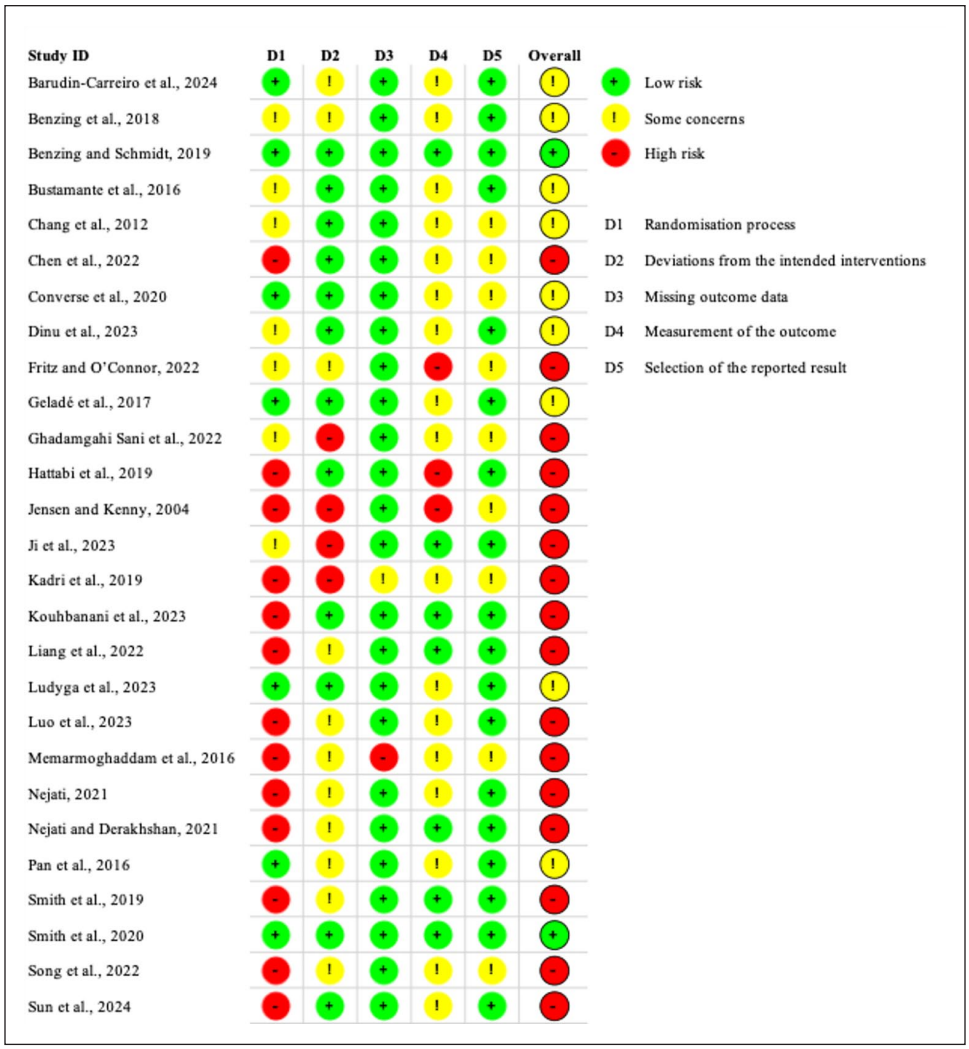


Figure 2. Methodological quality of included RCTs.

et al., 2024; Ludyga et al., 2017; Walters et al., 2024), while the remaining studies focused on chronic exercise interventions. For outcome measures, 7 studies assessed hyperactivity/impulsivity symptom (Benzing & Schmidt, 2019; Bustamante et al., 2016; Chen et al., 2022; Converse et al., 2020; Jensen & Kenny, 2004; Luo et al., 2023; F. Sun et al., 2024). Behavioral measures of Inhibitory control were reported in 34 studies (Barudin-Carreiro et al., 2024; Benzing et al., 2018; Benzing & Schmidt, 2019; Bigelow et al., 2021; Bustamante et al., 2016; Chang et al., 2012, 2014; Chen et al., 2022; Converse et al., 2020; Dinu et al., 2023; K. M. Fritz & O'Connor, 2016; K. Fritz & O'Connor, 2022; Geladé et al., 2017; Ghadamgahi Sani et al., 2022; Hattabi et al., 2019; Ji et al., 2023; Kadri et al., 2019; Kouhbanani et al., 2023; Kuo, Nitsche, et al., 2024; Kuo, Sun, et al., 2024; Li et al., 2024; Liang et al., 2022; Ludyga et al., 2017, 2023; Memarmoghaddam et al., 2016; Nejati, 2021; Nejati & Derakhshan, 2021; Pan et al., 2016, 2019; S. D. Smith et al., 2019, 2020; Y. Song et al., 2022; F. Sun et al., 2024; Walters et al., 2024). Regarding ERP measures, three studies examined N2 components (Ji et al., 2023; Ludyga et al., 2023; S. D. Smith et al., 2019), and another three studies focused on P3 components (Ludyga et al., 2017, 2023; S. D. Smith et al., 2019).

Risk of Bias Assessment

Figure 2 shows the risk of bias for RCTs, Figure 3 for crossover studies, and Table 2 for non-RCT studies. Among RCTs, 2 studies had a low risk of bias (Benzing & Schmidt, 2019; S. D. Smith et al., 2020), 9 had some concerns (Barudin-Carreiro et al., 2024; Benzing et al., 2018; Bustamante et al., 2016; Chang et al., 2012; Converse et al., 2020; Dinu et al., 2023; Geladé et al., 2017; Ludyga et al., 2023; Pan et al., 2016), and 16 had a high risk of bias (Chen et al., 2022; K. Fritz & O'Connor,

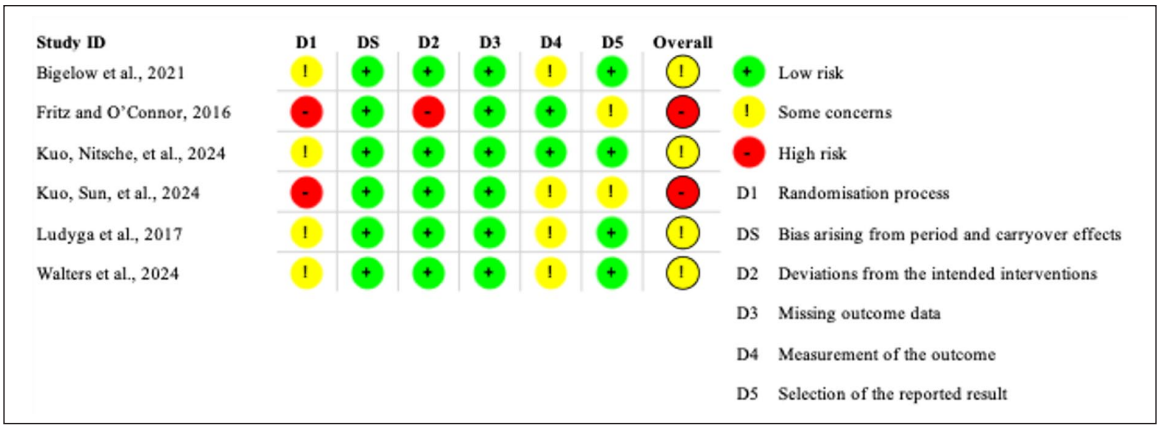


Figure 3. Methodological quality of included CRDs.

Table 2. ROBINS-I Assessment for the Methodological Quality of the Included non-RCT Studies.

Study	Confounding	Recruitment of participants	Classification of intervention	Deviation from intended intervention	Missing data	Measurement of outcomes	Selection of the reported result	Total
Chang et al. (2014)	Moderate risk	Low risk	Low risk	Low risk	Low risk	Moderate risk	Low risk	Moderate risk
Li et al. (2024)	Serious risk	Low risk	Low risk	Low risk	Low risk	Moderate risk	Low risk	Serious risk
Pan et al. (2019)	Moderate risk	Low risk	Low risk	Low risk	Low risk	Moderate risk	Low risk	Moderate risk

2022; Ghadamgahi Sani et al., 2022; Hattabi et al., 2019; Jensen & Kenny, 2004; Ji et al., 2023; Kadri et al., 2019; Kouhbanani et al., 2023; Liang et al., 2022; Luo et al., 2023; Memarmoghaddam et al., 2016; Nejati, 2021; Nejati & Derakhshan, 2021; S. D. Smith et al., 2019; Y. Song et al., 2022; F. Sun et al., 2024). In crossover studies, four had some concerns (Bigelow et al., 2021; Kuo, Nitsche, et al., 2024; Ludyga et al., 2017; Walters et al., 2024), while two had a high risk of bias (K. M. Fritz & O'Connor, 2016; Kuo, Sun, et al., 2024). Among non-RCTs, two had a moderate risk of bias (Chang et al., 2014; Pan et al., 2019), and one had a serious risk (Li et al., 2024).

Meta-Analysis of the Effects of Exercise on Hyperactivity/Impulsivity

A total of 7 studies, including 10 comparisons, evaluated hyperactivity/impulsivity symptoms as an outcome (Benzing & Schmidt, 2019; Bustamante et al., 2016; Chen et al., 2022; Converse et al., 2020; Jensen & Kenny, 2004; Luo et al., 2023; F. Sun et al., 2024). The results indicated that exercise interventions had no significant effect on hyperactivity/impulsivity symptoms in individuals with ADHD compared to control conditions ($g=-0.61$, 95% CI [−1.27, 0.06], $p=0.222$), with high heterogeneity ($I^2=83.8\%$, $p<.001$), as shown in Figure 4.

Meta-Analysis of the Effects of Exercise on Behavioral Performance of Inhibitory Control

A total of 34 studies, comprising 38 comparisons, evaluated the effect of exercise interventions on inhibitory control (Barudin-Carreiro et al., 2024; Benzing et al., 2018; Benzing & Schmidt, 2019; Bigelow et al., 2021; Bustamante et al., 2016; Chang et al., 2012, 2014; Chen et al., 2022; Converse et al., 2020; Dinu et al., 2023; K. M. Fritz & O'Connor, 2016; K. Fritz & O'Connor, 2022; Geladé et al., 2017; Ghadamgahi Sani et al., 2022; Hattabi et al., 2019; Ji et al., 2023; Kadri et al., 2019; Kouhbanani et al., 2023; Kuo, Nitsche, et al., 2024; Kuo, Sun, et al., 2024; Li et al., 2024; Liang et al., 2022; Ludyga et al., 2017, 2023; Memarmoghaddam et al., 2016; Nejati, 2021; Nejati & Derakhshan, 2021; Pan et al., 2016, 2019; S. D. Smith et al., 2019, 2020; Y. Song et al., 2022; F. Sun et al., 2024; Walters et al., 2024). The results showed that, compared to the control condition, exercise interventions had a significant but small-to-moderate effect on inhibitory control ($g=-0.47$, 95% CI [−0.71, −0.23], $p<.001$), with high heterogeneity ($I^2=79.3\%$, $p<.001$), as shown in Figure 5.

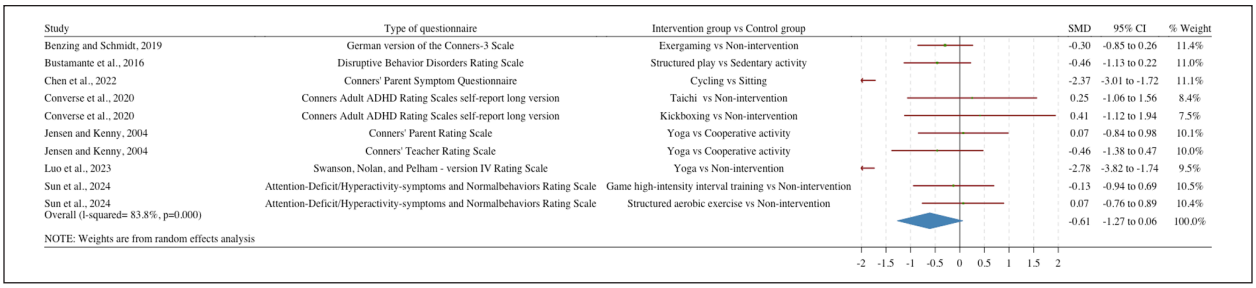


Figure 4. Forest plot of the meta-analysis of hyperactivity/impulsivity.

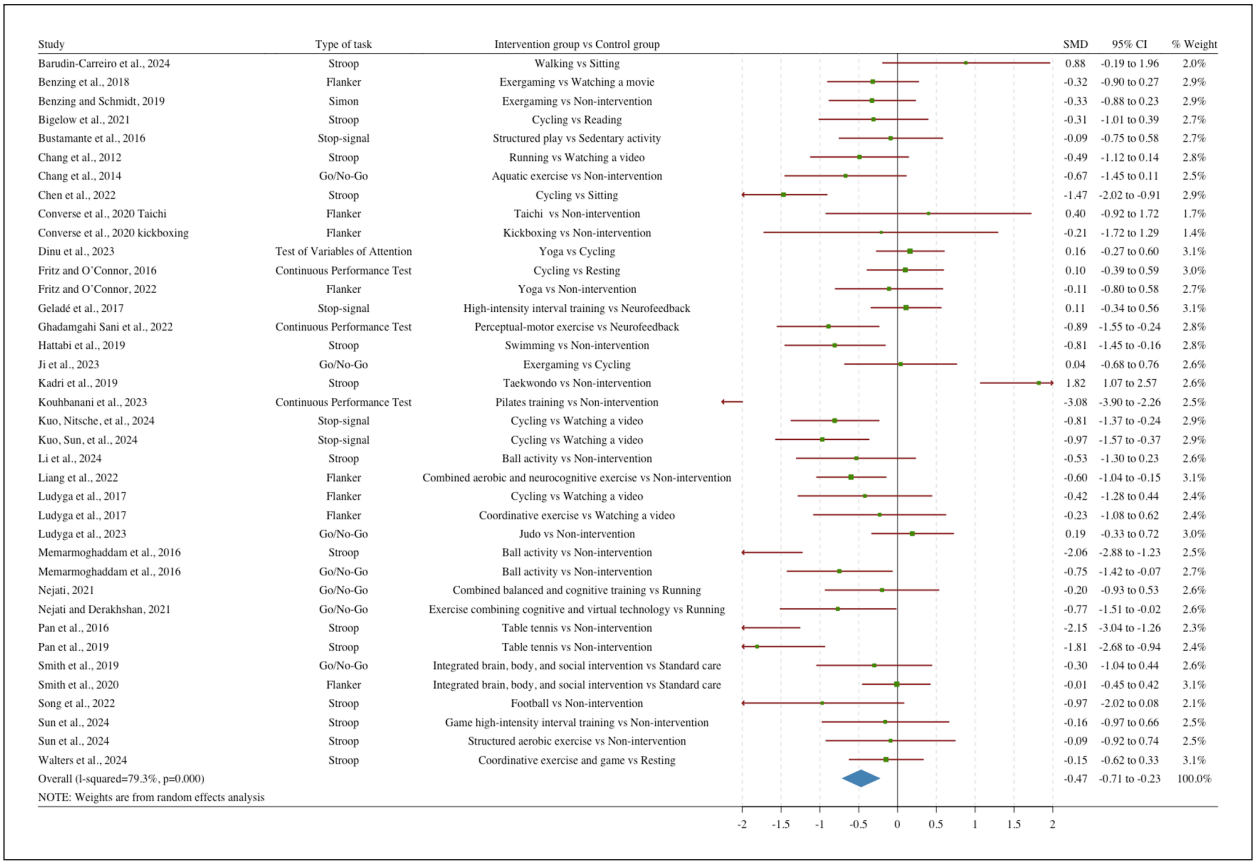


Figure 5. Forest plot of the meta-analysis of behavioral performance of inhibitory control.

Meta-Analysis of the Effects of Exercise on ERPs of Inhibitory Control

Three studies evaluated the N2 component as an outcome measure (Ji et al., 2023; Ludyga et al., 2023; S. D. Smith et al., 2019). The results indicated that, compared to the control condition, exercise interventions had no significant effect on the N2 component in individuals with ADHD ($g = -0.19$, 95% CI [−0.92, 0.54], $p = 0.610$), with high heterogeneity ($I^2 = 72.1\%$, $p = 0.028$), as shown in Figure 6. Additionally, four comparisons from three studies evaluated the P3 component as an outcome measure (Ludyga et al., 2017, 2023; S. D. Smith et al., 2019). The results indicated that, compared to the control condition, exercise interventions had no significant effect on the P3 component in individuals with ADHD (g = 0.06, 95% CI [−0.35, 0.47], $p = 0.789$), with better homogeneity ($I^2 = 22.8\%$, $p = 0.274$), as shown in Figure 7.

Subgroup Analysis

Due to the limited number of studies on hyperactivity/impulsivity symptoms and EEG studies on inhibitory control, subgroup analyses were not conducted. Instead, moderator

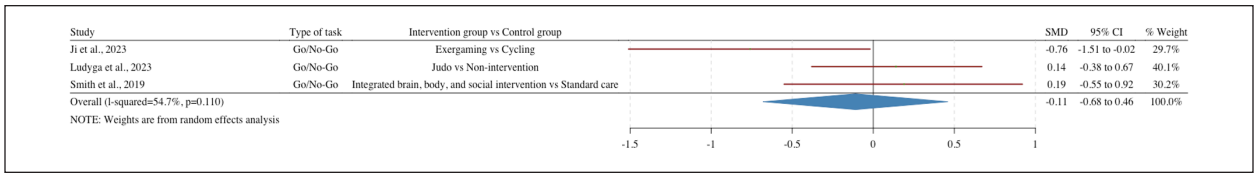


Figure 6. Forest plot of the meta-analysis of the N2.

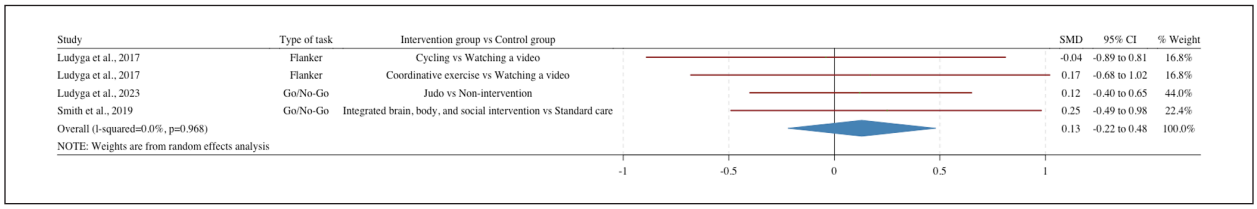


Figure 7. Forest plot of the meta-analysis of the P3.

variable analysis focused on behavioral performance of inhibitory control, considering subject age, inhibitory control domains, study design, exercise intervention type, exercise time, frequency, duration, intensity, motor skills, control group type, and study quality. In age subgroup analysis, exercise intervention showed a small-to-moderate effect in children and adolescents with ADHD ($g=-0.46$, 95% CI [−0.70, −0.21], $p<.001$) but no significant effect in adults ($g=-0.52$, 95% CI [−1.31, 0.27], $p=0.196$). Chronic interventions had a moderate-to-large effect ($g=-0.57$, 95% CI [−0.89, −0.24], $p=.001$), while acute interventions had no significant effect ($g=-0.26$, 95% CI [−0.52, 0.00], $p=.051$). Exercise frequency of three times per week showed a significant moderate-to-large effect ($g=-0.76$, 95% CI [−1.17, −0.35], $p<.001$), while twice ($g=-0.30$, 95% CI [−0.93, 0.33], $p=0.351$) and five times per week ($g=-0.43$, 95% CI [−1.27, 0.41], $p=0.317$) showed no significant effects. Exercise interventions had a moderate-to-large effect compared to sedentary/no-intervention groups ($g=-0.52$, 95% CI [−0.79, −0.24], $p<.001$) but did not differ from regular physical activity ($g=-0.13$, 95% CI [−0.52, 0.27], $p=0.537$) or neurofeedback training ($g=-0.36$, 95% CI [−1.35, 0.63], $p=0.473$). Subgroup analysis by study quality revealed a significant effect for moderate ($g=-0.34$, 95% CI [−0.64, −0.05], $p=.023$) and high-risk bias studies ($g=-0.61$, 95% CI [−1.01, −0.21], $p=.003$), but not for low-risk studies ($g=-0.13$, 95% CI [−0.48, 0.21], $p=0.445$). However, no moderating effects were observed for inhibitory control domains, study design, exercise time (min/session), exercise duration (weeks), exercise intensity, or motor skills. Detailed results are presented in Figure 8.

Sensitivity Analysis

Sensitivity analysis using the leave-one-out method revealed that the 95% confidence intervals of the pooled effect sizes for hyperactivity/impulsivity symptoms, behavioral performance of inhibitory control, and the N2 and P3 components of ERP remained largely unchanged, indicating that no single study substantially influenced the overall results.

Publication Bias

A visual inspection of the funnel plots (Figure 9(A) for hyperactivity/impulsivity, 9(B) for behavioral performance of inhibitory control, 9(C) for the N2 component, and 9(D) for the P3 component) suggested potential asymmetry. However, Egger's linear regression analysis revealed no statistical evidence of publication bias for behavioral performance of inhibitory control ($p=0.167$). Egger's tests were not conducted for hyperactivity/impulsivity, N2, and P3 due to the inclusion of fewer than 10 studies in each case.

Discussion

To our knowledge, this is the first systematic review and meta-analysis evaluating the effects of exercise interventions on inhibitory control in ADHD across three levels: questionnaire assessments, behavior, and electrophysiology. Synthesizing 36 studies (38 comparisons), we found a small to moderate significant effect of exercise on behavior performance of inhibitory control, supporting its efficacy in ADHD management. No significant effects were found on hyperactivity/impulsivity or inhibitory control-related ERP components.

Hyperactivity/Impulsivity Symptoms

Since lower symptom scores indicate improvement, negative effect sizes reflect benefits. Our meta-analysis found no

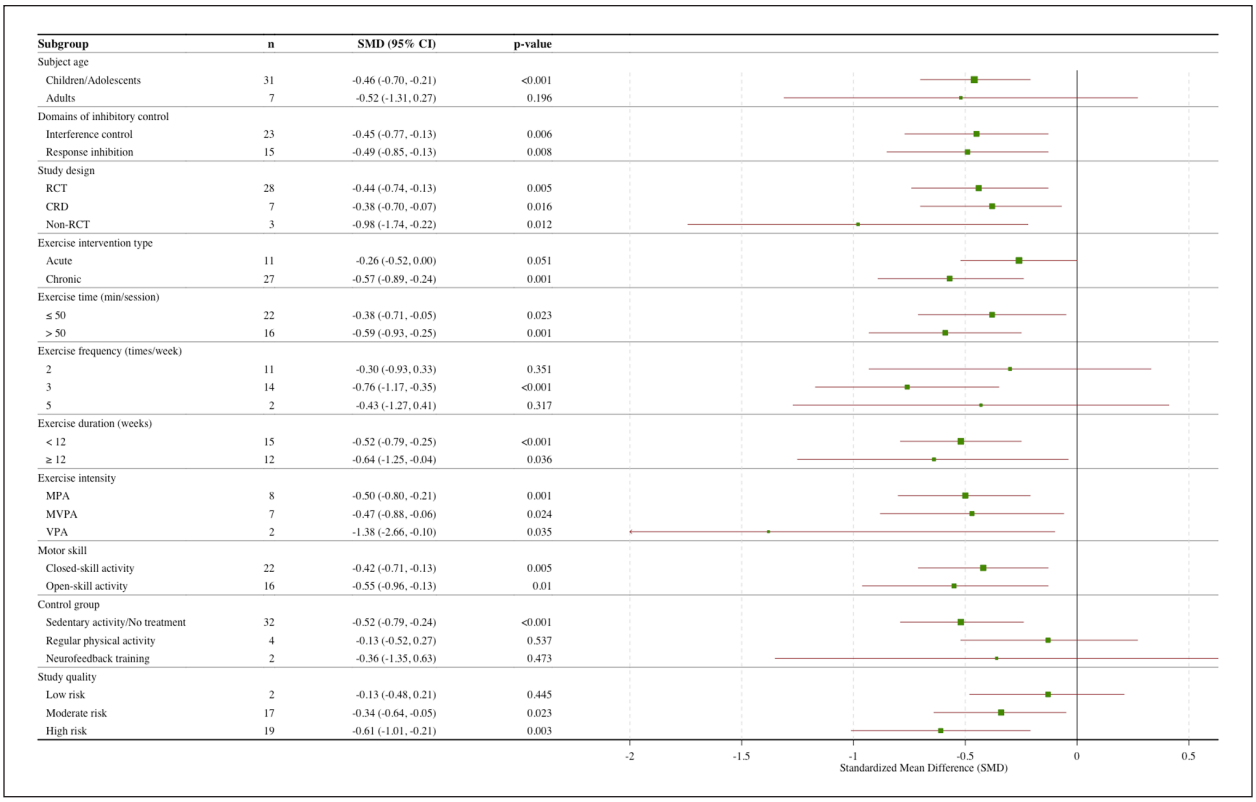


Figure 8. Subgroup analysis of behavioral performance of inhibitory control.

significant effects of exercise on hyperactivity/impulsivity, consistent with findings from studies focusing on children with ADHD (2022), but contrasting with Xie et al. (2021), who included both children and adults, highlighting the need for further investigation. The forest plot shows some effect sizes on the right of the null line, potentially contributing to the non-significant result. Possible reasons include small sample sizes (Converse et al., 2020; Jensen & Kenny, 2004), reducing statistical power and increasing variability. Sun et al. (2024) noted that studies conducted during the COVID-19 pandemic faced intervention period restrictions due to public health policies, potentially reducing intervention efficacy. Although our overall analysis did not detect significant effects, individual studies observed positive impacts of exercise on hyperactivity/impulsivity symptoms, necessitating further research to clarify these findings.

Behavioral Outcomes of Inhibitory Control

Our findings align with previous studies (H. Huang et al., 2022; Liang et al., 2021; Welsch et al., 2021), suggesting that exercise enhances inhibitory control in individuals with ADHD. In addition, a recent umbrella review also reported that exercise significantly improves inhibitory control in children with ADHD, with this conclusion supported by a high level of evidence (Dastamooz et al., 2023). Previous research mainly examined the effects through moderator analyses, whereas our study focused on inhibitory control as a single dimension and included more studies. Sensitivity analysis confirmed the robustness of our results, with no evidence of publication bias. Several moderators were identified, including participant age, exercise type, exercise frequency, control group type, and study quality.

Subgroup analysis showed exercise interventions were not significantly effective in adults, likely due to more stable inhibitory control from neurodevelopmental maturation. In contrast to children and adolescents, adults demonstrate lower impulsivity scores in tasks like CPT-AX, reflecting superior inhibitory control. Additionally, No-Go N2 amplitude declines with age (Jonkman, 2006), suggesting that adults may have already attained mature inhibitory control, influencing exercise intervention effectiveness. Nevertheless, this finding should be interpreted with caution, as it may be partly attributable to the small number of studies included in the adult subgroup. Acute exercise interventions did not show significant effects, possibly due to their transient impact on neural processes rather than sustained behavioral performance (Mehren et al., 2019). While acute exercise can enhance cerebral blood flow (Mulser & Moreau, 2023) and brain-derived neurotrophic factor (Borror, 2017), these

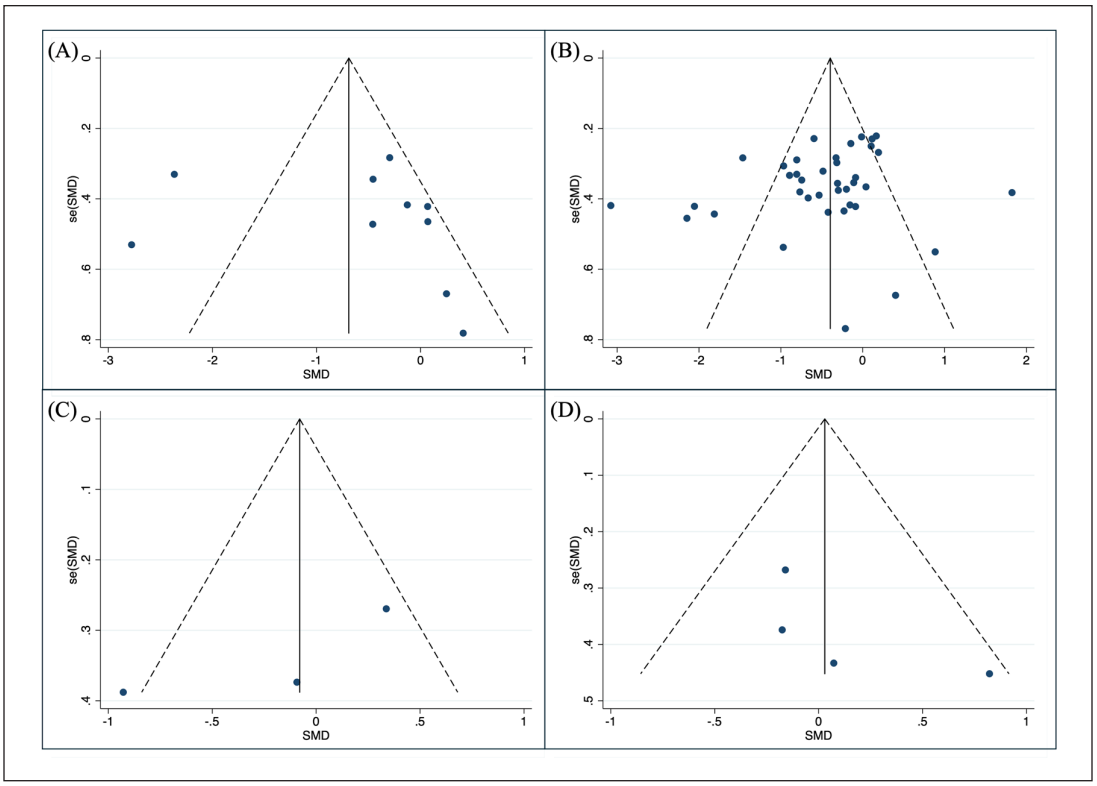


Figure 9. Funnel plot for hyperactivity/impulsivity (A), behavioral performance of inhibitory control (B), the N2 (C), and the P3 (D).

short-term changes may not yield lasting behavioral improvements. Chronic exercise, on the other hand, also increases gray matter volume (Tarkka et al., 2019), optimizes neurophysiological functioning (Meijer et al., 2020), and modulates neurotransmitter systems (Chan et al., 2022), leading to more substantial and sustained improvements in inhibitory control. Three sessions per week produced significant effects, supporting the inverted-U hypothesis, which suggests moderate intervention frequency optimizes outcomes. Both insufficient and excessive frequencies may hinder efficacy. Exercise interventions did not outperform regular physical activity and neurofeedback training, both of which have been shown to enhance inhibitory control. However, exercise interventions showed more significant benefits when compared to passive control groups (e.g., sedentary conditions or no-treatment controls). Studies with a low risk of bias did not demonstrate significant improvements. Notably, the low-bias-risk category included two RCTs that implemented rigorous randomization and blinding procedures, minimizing expectation effects from both researchers and participants. Consequently, effect size estimates in these studies were likely more objective and conservative. In contrast, studies with moderate or high risk of bias, which exhibited methodological limitations such as inadequate control conditions or randomization procedures, tended to report larger effect sizes. No moderating effects were found for inhibitory control domains, study design, exercise duration per session, total intervention duration, exercise intensity, or motor skill level.

Electrophysiological Measures of Inhibitory Control

The N2 component, a negative deflection linked to conflict monitoring, reflects response inhibition in the Go/No-Go task (Groom & Cragg, 2015; Randall & Smith, 2011). A more negative value indicates greater improvement. Among the three studies, one (Ji et al., 2023) showed an effect size on the right side of the null line, comparing exergaming and cycling. While no significant pre-post changes in No-Go N2 amplitude were found, the exergaming group showed a significant increase in Go N2 amplitude, suggesting that exergaming may enhance response execution rather than directly improving response inhibition. The P3 component, a positive deflection associated with inhibitory control (Gang & Tiejun, 2010; Groom & Cragg, 2015), attentional resource allocation, and context updating (Polich, 2007), serves as an index where greater positive values indicate improved performance. Three studies (four comparisons) used Go/No-Go and Flanker tasks. Two studies (Ludyga et al., 2023; S. D. Smith et al., 2019) reported effect sizes to the left of the null line. In the study by Ludyga et al. (2023), the judo intervention did not yield significant improvements, which may be attributed to the participants' relatively low motor skill levels.

Motor proficiency is foundational for mastering complex movement techniques, and children with ADHD who exhibit lower motor skills may experience heightened cognitive demands when learning judo, potentially diminishing intervention benefits. Similarly, S. D. Smith et al. (2019) observed no significant effect of exercise on P3, possibly due to the study's small sample size and limited intervention dosage. Although no significant effects of exercise interventions on N2 and P3 components were observed, these findings do not rule out their potential benefits for inhibitory control. Future research should use neuroimaging to explore the neural mechanisms underlying exercise-induced improvements.

Limitations

This study has several limitations: (1) Many studies reported only total symptom scores, limiting subgroup analyses on hyperactivity/impulsivity; (2) The small number of EEG studies may have reduced our ability to detect significant effects, and future research should integrate neuroimaging techniques for more comprehensive evidence; (3) Few studies were classified as low-risk, and future RCTs should follow CONSORT guidelines for greater transparency in study design, randomization, and blinding; (4) Some studies lacked detailed diagnostic criteria and intervention parameters, limiting subgroup analyses. Future studies should provide comprehensive methodological details for better reproducibility and comparability.

Conclusions

This meta-analysis shows a small-to-moderate beneficial effect of exercise on inhibitory control in ADHD, significantly moderated by age, intervention type, frequency, control group type, and study quality. However, exercise did not show significant effects on hyperactivity, impulsivity, or inhibition-related N2 and P3 components, suggesting that its impact should not be overestimated. These findings underscore the necessity of implementing chronic exercise interventions and maintaining a moderate exercise frequency from childhood to promote the development of inhibitory control. Future RCTs employing neuroimaging techniques and following CONSORT guidelines are needed to clarify the mechanisms of exercise and verify its effectiveness in ADHD management.

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Ethical Considerations

The protocol for this study was registered with PROSPERO (CRD42025637758).

Author Contributions

Zeping Zhang, Yongfei Zhu, and Suyong Yang developed the idea of systematic review and meta-analysis. Literature search was conducted by Xuanyu Bo and Kun Liu. Literature screening and data extraction were performed by Xuanyu Bo and Kun Liu. Data analysis was conducted by Zeping Zhang and Jiangdi Su. The first draft of the manuscript was written by Zeping Zhang, and all the authors made critical modifications to the manuscript. All authors read and approved the final manuscript.

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Declaration of Conflicting Interests

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Supplemental Material

Supplemental material for this article is available online.

References

American Psychiatric Association. (2022). *Diagnostic and statistical manual of mental disorders (DSM-5 TR)* (5th ed., text revision ed.).

Ayano, G., Demelash, S., Gizachew, Y., Tsegay, L., & Alati, R. (2023). The global prevalence of attention deficit hyperactivity disorder in children and adolescents: An umbrella review of meta-analyses. *Journal of Affective Disorders*, *339*, 860–866. https://doi.org/10.1016/j.jad.2023.07.071

Bari, A., & Robbins, T. W. (2013). Inhibition and impulsivity: Behavioral and neural basis of response control. *Progress in Neurobiology*, *108*, 44–79. https://doi.org/10.1016/j.pneurobio.2013.06.005

Barkley, R. A. (1997). Behavioral inhibition, sustained attention, and executive functions: Constructing a unifying theory of ADHD. *Psychological Bulletin*, *121*(1), 65–94. https://doi.org/10.1037/0033-2909.121.1.65

Barudin-Carreiro, A., Camhi, S. M., Stanish, H. I., & Wright, J. A. (2024). Acute effects of walking and standing on executive function in children with Attention Deficit/Hyperactivity Disorder: A feasibility study. *Children*, *11*(3), 341. https://doi.org/10.3390/children11030341

Benzing, V., Chang, Y.-K., & Schmidt, M. (2018). Acute physical activity enhances executive functions in children with ADHD. *Scientific Reports*, *8*(1), 12382. https://doi.org/10.1038/s41598-018-30067-8

Benzing, V., & Schmidt, M. (2019). The effect of exergaming on executive functions in children with ADHD: A randomized clinical trial. *Scandinavian Journal of Medicine & Science in Sports*, *29*(8), 1243–1253. https://doi.org/10.1111/sms.13446

Berwid, O. G., & Halperin, J. M. (2012). Emerging support for a role of exercise in Attention-Deficit/Hyperactivity Disorder

intervention planning. *Current Psychiatry Reports*, *14*(5), 543–551. https://doi.org/10.1007/s11920-012-0297-4

Bigelow, H., Gottlieb, M. D., Ogrodnik, M., Graham, J. D., & Fenesi, B. (2021). The differential impact of acute exercise and mindfulness meditation on executive functioning and psycho-emotional well-being in children and youth with ADHD. *Frontiers in Psychology*, *12*, 660845. https://doi.org/10.3389/fpsyg.2021.660845

Borror, A. (2017). Brain-derived neurotrophic factor mediates cognitive improvements following acute exercise. *Medical Hypotheses*, *106*, 1–5. https://doi.org/10.1016/j.mehy.2017.06.024

Broyd, S. J., Johnstone, S. J., Barry, R. J., Clarke, A. R., McCarthy, R., Selikowitz, M., & Lawrence, C. A. (2005). The effect of methylphenidate on response inhibition and the event-related potential of children with Attention Deficit/Hyperactivity Disorder. *International Journal of Psychophysiology*, *58*(1), 47–58. https://doi.org/10.1016/j.ijpsycho.2005.03.008

Bustamante, E. E., Davis, C. L., Frazier, S. L., Rusch, D., Fogg, L. F., Atkins, M. S., & Marquez, D. X. (2016). Randomized controlled trial of exercise for ADHD and disruptive behavior disorders. *Medicine & Science in Sports & Exercise*, *48*(7), 1397–1407. https://doi.org/10.1249/MSS.0000000000000891

Chan, Y.-S., Jang, J.-T., & Ho, C.-S. (2022). Effects of physical exercise on children with attention deficit hyperactivity disorder. *Biomedical Journal*, *45*(2), 265–270. https://doi.org/10.1016/j.bj.2021.11.011

Chang, Y.-K., Hung, C.-L., Huang, C.-J., Hatfield, B. D., & Hung, T.-M. (2014). Effects of an aquatic exercise program on inhibitory control in children with ADHD: A preliminary study. *Archives of Clinical Neuropsychology*, *29*(3), 217–223. https://doi.org/10.1093/arclin/acu003

Chang, Y.-K., Liu, S., Yu, H.-H., & Lee, Y.-H. (2012). Effect of acute exercise on executive function in children with attention deficit hyperactivity disorder. *Archives of Clinical Neuropsychology*, *27*(2), 225–237. https://doi.org/10.1093/arclin/acr094

Chen, X., Liang, G., Li, M., Wang, Q., & Su, M. (2022). Effect of aerobic exercise on core symptoms and executive function in children with attention deficit hyperactivity disorder. *Chinese Journal of Rehabilitation Theory and Practice*, *6*, 704–709.

Chmielewski, W. X., Tiedt, A., Bluschke, A., Dippel, G., Roessner, V., & Beste, C. (2018). Effects of multisensory stimuli on inhibitory control in adolescent ADHD: It is the content of information that matters. *NeuroImage: Clinical*, *19*, 527–537. https://doi.org/10.1016/j.nicl.2018.05.019

The Cochrane Collaboration. (2021). *Revised Cochrane risk of bias tool for randomised trials (RoB 2) Additional considerations for crossover trials*. https://sites.google.com/site/riskofbiastool/welcome/rob-2-0-tool/rob-2-for-crossover-trials

Converse, A. K., Barrett, B. P., Chewning, B. A., & Wayne, P. M. (2020). Tai Chi training for attention deficit hyperactivity disorder: A feasibility trial in college students. *Complementary Therapies in Medicine*, *53*, Article 102538. https://doi.org/10.1016/j.ctim.2020.102538

Dastamooz, S., Sadeghi-Bahmani, D., Farahani, M. H. D., Wong, S. H. S., Yam, J. C. S., Tham, C. C. Y., & Sit, C. H. P. (2023). The efficacy of physical exercise interventions on mental health, cognitive function, and ADHD symptoms in children and adolescents with ADHD: An umbrella review. *EClinicalMedicine*, *62*, Article 102137. https://doi.org/10.1016/j.eclinm.2023.102137

Diamond, A. (2013). Executive functions. *Annual Review of Psychology*, *64*(1), 135–168. https://doi.org/10.1146/annurev-psych-113011-143750

Dinu, L. M., Singh, S. N., Baker, N. S., Georgescu, A. L., Singer, B. F., Overton, P. G., & Dommett, E. J. (2023). The effects of different exercise approaches on attention deficit hyperactivity disorder in adults: A randomised controlled trial. *Behavioral Sciences*, *13*(2), Article 2. https://doi.org/10.3390/bs13020129

Doehnert, M., Brandeis, D., Imhof, K., Drechsler, R., & Steinhausen, H.-C. (2010). Mapping Attention-Deficit/Hyperactivity Disorder from childhood to adolescence—No neurophysiologic evidence for a developmental lag of attention but some for inhibition. *Biological Psychiatry*, *67*(7), 608–616. https://doi.org/10.1016/j.biopsych.2009.07.038

Fritz, K. M., & O'Connor, P. J. (2016). Acute exercise improves mood and motivation in young men with ADHD symptoms. *Medicine and Science in Sports and Exercise*, *48*(6), 1153–1160. https://doi.org/10.1249/MSS.0000000000000864

Fritz, K., & O'Connor, P. J. (2022). Effects of a 6 week yoga intervention on executive functioning in women screening positive for adult ADHD: A pilot study. *Front Sports Act Living*, *4*, Article 746409. https://doi.org/10.3389/fspor.2022.746409

Gang, Y., & Tie-jun, L. (2010, April 16–18). *Neural mechanism of response inhibition in a Go/NoGo tasks* [Conference session]. 2010 2nd IEEE International Conference on Information Management and Engineering, Chengdu, China (pp. 498–503). https://doi.org/10.1109/ICIME.2010.5478126

Geladé, K., Bink, M., Janssen, T. W. P., Van Mourik, R., Maras, A., & Oosterlaan, J. (2017). An RCT into the effects of neurofeedback on neurocognitive functioning compared to stimulant medication and physical activity in children with ADHD. *European Child & Adolescent Psychiatry*, *26*(4), 457–468. https://doi.org/10.1007/s00787-016-0902-x

Ghadamgahi Sani, N., Akbarfahimi, M., Akbari, S., Alizadeh Zarei, M., & Taghizadeh, G. (2022). Neurofeedback training versus perceptual-motor exercises interventions in visual attention for children with attention-deficit/hyperactivity disorder: A randomized controlled trial. *Basic and Clinical Neuroscience*, *13*(2), 215–224. https://doi.org/10.32598/bcn.2021.563.2

Groom, M. J., & Cragg, L. (2015). Differential modulation of the N2 and P3 event-related potentials by response conflict and inhibition. *Brain and Cognition*, *97*, 1–9. https://doi.org/10.1016/j.bandc.2015.04.004

Hattabi, S., Bouallegue, M., Ben Yahya, H., & Bouden, A. (2019). Rehabilitation of ADHD children by sport intervention: A Tunisian experience. *La Tunisie Médicale*, *97*(7), 874–881.

Hedges, L. V. (1981). Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, *6*(2), 107–128.

Huang, H., Jin, Z., He, C., Guo, S., Zhang, Y., & Quan, M. (2022). Chronic exercise for core symptoms and executive functions

in ADHD: A meta-analysis. *Pediatrics*, *151*(1), e2022057745. https://doi.org/10.1542/peds.2022-057745

Huang, K.-L., Wei, H.-T., Hsu, J.-W., Bai, Y.-M., Su, T.-P., Li, C.-T., Lin, W.-C., Tsai, S.-J., Chang, W.-H., Chen, T.-J., & Chen, M.-H. (2018). Risk of suicide attempts in adolescents and young adults with attention-deficit hyperactivity disorder: A nationwide longitudinal study. *British Journal of Psychiatry*, *212*(4), 234–238. https://doi.org/10.1192/bjp.2018.8

Jangmo, A., Stålhandske, A., Chang, Z., Chen, Q., Almqvist, C., Feldman, I., Bulik, C. M., Lichtenstein, P., D'Onofrio, B., Kuja-Halkola, R., & Larsson, H. (2019). Attention-deficit/hyperactivity disorder, school performance, and effect of medication. *Journal of the American Academy of Child & Adolescent Psychiatry*, *58*(4), 423–432. https://doi.org/10.1016/j.jaac.2018.11.014

Jensen, P. S., & Kenny, D. T. (2004). The effects of yoga on the attention and behavior of boys with Attention-Deficit/hyperactivity Disorder (ADHD). *Journal of Attention Disorders*, *7*(4), 205–216. https://doi.org/10.1177/108705470400700403

Ji, H., Wu, S., Won, J., Weng, S., Lee, S., Seo, S., & Park, J.-J. (2023). The effects of exergaming on attention in children with Attention Deficit/Hyperactivity Disorder: Randomized controlled trial. *JIMR Serious Games*, *11*, e40438. https://doi.org/10.2196/40438

Johnstone, S. J., Barry, R. J., Markovska, V., Dimoska, A., & Clarke, A. R. (2009). Response inhibition and interference control in children with AD/HD: A visual ERP investigation. *International Journal of Psychophysiology*, *72*(2), 145–153. https://doi.org/10.1016/j.ijpsycho.2008.11.007

Johnstone, S. J., & Clarke, A. R. (2009). Dysfunctional response preparation and inhibition during a visual Go/Nogo task in children with two subtypes of attention-deficit hyperactivity disorder. *Psychiatry Research*, *166*(2–3), 223–237. https://doi.org/10.1016/j.psychres.2008.03.005

Jonkman, L. M. (2006). The development of preparation, conflict monitoring and inhibition from early childhood to young adulthood; A Go/Nogo ERP study. *Brain Research*, *1097*(1), 181–193. https://doi.org/10.1016/j.brainres.2006.04.064

Kadri, A., Slimani, M., Bragazzi, N., Tod, D., & Azaiez, F. (2019). Effect of Taekwondo practice on cognitive function in adolescents with Attention Deficit Hyperactivity Disorder. *International Journal of Environmental Research and Public Health*, *16*(2), 204. https://doi.org/10.3390/ijerph16020204

Kouhbanani, S. S., Zarenezhad, S., & Arabi, S. M. (2023). Mind-body exercise affects attention switching and sustained attention in female adults with Attention Deficit/Hyperactivity Disorder: A randomized, controlled trial with 6-month follow-up. *Current Psychology*, *42*(24), 20983–20994. https://doi.org/10.1007/s12144-022-03216-6

Kuo, H.-I., Nitsche, M. A., Wu, Y.-T., Chang, J.-C., & Yang, L.-K. (2024). Acute aerobic exercise modulates cognition and cortical excitability in adults with attention-deficit hyperactivity disorder (ADHD) and healthy controls. *Psychiatry Research*, *340*, Article 116108. https://doi.org/10.1016/j.psychres.2024.116108

Kuo, H.-I., Sun, J.-L., Nitsche, M., & Chang, J.-C. (2024). An investigation of the acute effects of aerobic exercise on executive function and cortical excitability in adolescents with attention deficit hyperactivity disorder (ADHD). *European Child & Adolescent Psychiatry*, *33*(12), 4169–4183. https://doi.org/10.1007/s00787-024-02467-x

Lee, S. S., Humphreys, K. L., Flory, K., Liu, R., & Glass, K. (2011). Prospective association of childhood attention-deficit/hyperactivity disorder (ADHD) and substance use and abuse/dependence: A meta-analytic review. *Clinical Psychology Review*, *31*(3), 328–341. https://doi.org/10.1016/j.cpr.2011.01.006

Li, L., Huang, Z., Yang, Y., & Wu, X. (2024). Effect of ball-based physical activity on basic motor skills and executive function in children with attention deficit hyperactivity disorder. *Chinese Journal of Rehabilitation Theory and Practice*, *4*, 479–486.

Liang, X., Li, R., Wong, S. H. S., Sum, R. K. W., & Sit, C. H. P. (2021). The impact of exercise interventions concerning executive functions of children and adolescents with attention-deficit/hyperactive disorder: A systematic review and meta-analysis. *International Journal of Behavioral Nutrition and Physical Activity*, *18*(1), 68. https://doi.org/10.1186/s12966-021-01135-6

Liang, X., Qiu, H., Wang, P., & Sit, C. H. P. (2022). The impacts of a combined exercise on executive function in children with ADHD: A randomized controlled trial. *Scandinavian Journal of Medicine & Science in Sports*, *32*(8), 1297–1312. https://doi.org/10.1111/sms.14192

Liberati, A., Altman, D. G., Tetzlaff, J., Mulrow, C., Gøtzsche, P. C., Ioannidis, J. P., Clarke, M., Devereaux, P. J., Kleijnen, J., & Moher, D. (2009). The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: Explanation and elaboration. *BMJ*, *339*, b2700. https://doi.org/10.1136/bmj.b2700

Ludyga, S., Brand, S., Gerber, M., Weber, P., Brotzmann, M., Habibifar, F., & Pühse, U. (2017). An event-related potential investigation of the acute effects of aerobic and coordinative exercise on inhibitory control in children with ADHD. *Developmental Cognitive Neuroscience*, *28*, 21–28. https://doi.org/10.1016/j.dcn.2017.10.007

Ludyga, S., Hanke, M., Leuenberger, R., Bruggisser, F., Pühse, U., Gerber, M., Lemola, S., Capone-Mori, A., Keutler, C., Brotzmann, M., & Weber, P. (2023). Martial arts and cognitive control in children with Attention-Deficit Hyperactivity Disorder and children born very preterm: A combined analysis of two randomized controlled trials. *Medicine & Science in Sports & Exercise*, *55*(5), 777–786. https://doi.org/10.1249/MSS.0000000000003110

Luo, X., Huang, X., & Lin, S. (2023). Yoga and music intervention reduces inattention, hyperactivity/impulsivity, and oppositional defiant disorder in children's consumer with comorbid ADHD and ODD. *Frontiers in Psychology*, *14*, Article 1150018. https://doi.org/10.3389/fpsyg.2023.1150018

Marquardt, L., Eichele, H., Lundervold, A. J., Haavik, J., & Eichele, T. (2018). Event-related-potential (ERP) correlates of performance monitoring in adults with Attention-Deficit Hyperactivity Disorder (ADHD). *Frontiers in Psychology*, *9*, Article 485. https://www.frontiersin.org/articles/10.3389/fpsyg.2018.00485

Mehren, A., Özyurt, J., Thiel, C. M., Brandes, M., Lam, A. P., & Philipsen, A. (2019). Effects of acute aerobic exercise on response inhibition in adult patients with ADHD. *Scientific*

Reports, *9*(1), Article 19884. https://doi.org/10.1038/s41598-019-56332-y

Meijer, A., Konigs, M., Vermeulen, G. T., Visscher, C., Bosker, R. J., Hartman, E., Oosterlaan, J., & Oosterlaan, J. (2020). The effects of physical activity on brain structure and neurophysiological functioning in children: A systematic review and meta-analysis. *Developmental Cognitive Neuroscience*, *45*, Article 100828. https://doi.org/10.1016/j.dcn.2020.100828

Memarmoghaddam, M., Torbati, H., Sohrabi, M., Mashhadi, A., & Kashi, A. (2016). Effects of a selected exercise program on executive function of children with attention deficit hyperactivity disorder. *Journal of Medicine and Life*, *9*(4), 373–379.

Mulser, L., & Moreau, D. (2023). Effect of acute cardiovascular exercise on cerebral blood flow: A systematic review. *Brain Research*, *1809*, Article 148355. https://doi.org/10.1016/j.brainres.2023.148355

Nejati, V. (2021). Balance-based Attentive Rehabilitation of Attention Networks (BARAN) improves executive functions and ameliorates behavioral symptoms in children with ADHD. *Complementary Therapies in Medicine*, *60*, Article 102759. https://doi.org/10.1016/j.ctim.2021.102759

Nejati, V., & Derakhshan, Z. (2021). The effect of physical activity with and without cognitive demand on the improvement of executive functions and behavioral symptoms in children with ADHD. *Expert Review of Neurotherapeutics*, *21*(5), 607–614. https://doi.org/10.1080/14737175.2021.1912600

Nijmeijer, J. S., Minderaa, R. B., Buitelaar, J. K., Mulligan, A., Hartman, C. A., & Hoekstra, P. J. (2008). Attention-deficit/hyperactivity disorder and social dysfunctioning. *Clinical Psychology Review*, *28*(4), 692–708. https://doi.org/10.1016/j.cpr.2007.10.003

Norman, L. J., Carlisi, C., Lukito, S., Hart, H., Mataix-Cols, D., Radua, J., & Rubia, K. (2016). Structural and functional brain abnormalities in Attention-Deficit/Hyperactivity Disorder and obsessive-compulsive disorder a comparative meta-analysis. *JAMA Psychiatry*, *73*(8), 815–825. https://doi.org/10.1001/jamapsychiatry.2016.0700

Pan, C.-Y., Chu, C.-H., Tsai, C.-L., Lo, S.-Y., Cheng, Y.-W., & Liu, Y.-J. (2016). A racket-sport intervention improves behavioral and cognitive performance in children with attention-deficit/hyperactivity disorder. *Research in Developmental Disabilities*, *57*, 1–10. https://doi.org/10.1016/j.ridd.2016.06.009

Pan, C.-Y., Tsai, C.-L., Chu, C.-H., Sung, M.-C., Huang, C.-Y., & Ma, W.-Y. (2019). Effects of physical exercise intervention on motor skills and executive functions in children with ADHD: A pilot study. *Journal of Attention Disorders*, *23*(4), 384–397. https://doi.org/10.1177/1087054715569282

Polich, J. (2007). Updating P300: An integrative theory of P3a and P3b. *Clinical Neurophysiology*, *118*(10), 2128–2148. https://doi.org/10.1016/j.clinph.2007.04.019

Randall, W. M., & Smith, J. L. (2011). Conflict and inhibition in the cued-Go/NoGo task. *Clinical Neurophysiology*, *122*(12), 2400–2407. https://doi.org/10.1016/j.clinph.2011.05.012

Seiffer, B., Hautzinger, M., Ulrich, R., & Wolf, S. (2022). The efficacy of physical activity for children with Attention Deficit Hyperactivity Disorder: A meta-analysis of randomized controlled trials. *Journal of Attention Disorders*, *26*(5), 656–673. https://doi.org/10.1177/10870547211017982

Smith, J. L., Jamadar, S., Provost, A. L., & Michie, P. T. (2013). Motor and non-motor inhibition in the Go/NoGo task: An ERP and fMRI study. *International Journal of Psychophysiology*, *87*(3), 244–253.

Smith, J. L., Johnstone, S. J., & Barry, R. J. (2008). Movement-related potentials in the Go/NoGo task: The P3 reflects both cognitive and motor inhibition. *Clinical Neurophysiology*, *119*(3), 704–714. https://doi.org/10.1016/j.clinph.2007.11.042

Smith, S. D., Crowley, M. J., Ferrey, A., Ramsey, K., Wexler, B. E., Leckman, J. F., & Sukhodolsky, D. G. (2019). Effects of integrated brain, body, and social (IBBS) intervention on ERP measures of attentional control in children with ADHD. *Psychiatry Research*, *278*, 248–257. https://doi.org/10.1016/j.psychres.2019.06.021

Smith, S. D., Vitulano, L. A., Katsovich, L., Li, S., Moore, C., Li, F., Grantz, H., Zheng, X., Eicher, V., Aktan Guloksuz, S., et al. (2020). A randomized controlled trial of an integrated brain, body, and social intervention for children with ADHD. *Journal of Attention Disorders*, *24*(5), 780–794. https://doi.org/10.1177/1087054716647490

Song, P., Zha, M., Yang, Q., Zhang, Y., Li, X., & Rudan, I. (2021). The prevalence of adult attention-deficit hyperactivity disorder: A global systematic review and meta-analysis. *Journal of Global Health*, *11*, Article 04009. https://doi.org/10.7189/jogh.11.04009

Song, Y., Li, Y., Liu, J., Li, X., Yin, T., Wang, F., Lu, S., Chen, M., & Ren, Y. (2022). Effects of football exercise on executive function of boys with attention deficit hyperactivity disorder. *Chinese Journal of Sports Medicine*, *3*, 165–172. https://doi.org/10.16038/j.1000-6710.2022.03.010

Stern, A., Agnew-Blais, J. C., Danese, A., Fisher, H. L., Matthews, T., Polanczyk, G., Wertz, J., & Arseneault, L. (2020). Associations between ADHD and emotional problems from childhood to young adulthood: A longitudinal genetically sensitive study. *Journal of Child Psychology and Psychiatry*, *61*(11), 1234–1242. https://doi.org/10.1111/jcpp.13217

Sterne, J. A. C., Hernan, M. A., Reeves, B. C., Savovic, J., Berkman, N. D., Viswanathan, M., Henry, D., Altman, D. G., Ansari, M. T., Boutron, I., Carpenter, J. R., Chan, A.-W., Churchill, R., Deeks, J. J., Hrobjartsson, A., Kirkham, J., Juni, P., Loke, Y. K., Pigott, T. D., . . . Higgins, J. P. T. (2016). ROBINS-I: A tool for assessing risk of bias in non-randomised studies of interventions. *BMJ-British Medical Journal*, *355*, i4919. https://doi.org/10.1136/bmj.i4919

Sterne, J. A. C., Savović, J., Page, M. J., Elbers, R. G., Blencowe, N. S., Boutron, I., Cates, C. J., Cheng, H.-Y., Corbett, M. S., Eldridge, S. M., Emberson, J. R., Hernán, M. A., Hopewell, S., Hróbjartsson, A., Junqueira, D. R., Jüni, P., Kirkham, J. J., Lasserson, T., Li, T., . . . Higgins, J. P. T. (2019). RoB 2: A revised tool for assessing risk of bias in randomised trials. *BMJ*, *366*, l4898. https://doi.org/10.1136/bmj.l4898

Sun, F., Fang, Y., Ho, Y. F., Chow, G. C., Yang, Y., Huang, K., Yu, C. C., Liu, D., Wong, S. H., Siu, P. M., et al. (2024). Effectiveness of a game-based high-intensity interval training on executive function and other health indicators of children with ADHD: A three-arm partially-blinded randomized controlled trial. *Journal of Exercise Science and Fitness*, *22*(4), 408–416. https://doi.org/10.1016/j.jesf.2024.09.001

Sun, W., Yu, M., & Zhou, X. (2022). Effects of physical exercise on attention deficit and other major symptoms in children with ADHD: A meta-analysis. *Psychiatry Research*, *311*, Article 114509. https://doi.org/10.1016/j.psychres.2022.114509

Tarkka, I. M., Hautasaari, P., Pesonen, H., Niskanen, E., Rottensteiner, M., Kaprio, J., Savic, A. M., & Kujala, U. M. (2019). Long-term physical activity may modify brain structure and function: Studies in young healthy twins. *Journal of Physical Activity & Health*, *16*(8), 637–643. https://doi.org/10.1123/jpah.2018-0416

Uebel, H., Albrecht, B., Asherson, P., Börger, N. A., Butler, L., Chen, W., Christiansen, H., Heise, A., Kuntsi, J., Schäfer, U., Andreou, P., Manor, I., Marco, R., Miranda, A., Mulligan, A., Oades, R. D., Van Der Meere, J., Faraone, S. V., Rothenberger, A., & Banaschewski, T. (2010). Performance variability, impulsivity errors and the impact of incentives as gender-independent endophenotypes for ADHD. *Journal of Child Psychology and Psychiatry*, *51*(2), 210–218. https://doi.org/10.1111/j.1469-7610.2009.02139.x

Walters, G. W., Taylor, S., Sweeney, E. L., Cooper, S. B., Williams, R. A., & Dring, K. J. (2024). Effect of an acute bout of exercise on executive function and sleep in children with attention deficit hyperactivity disorder and autism spectrum disorder. *Mental Health and Physical Activity*, *27*. 1–17. https://doi.org/10.1016/j.mhpa.2024.100624

Welsch, L., Alliott, O., Kelly, P., Fawkner, S., Booth, J., & Niven, A. (2021). The effect of physical activity interventions on executive functions in children with ADHD: A systematic review and meta-analysis. *Mental Health and Physical Activity*, *20*, Article 100379. https://doi.org/10.1016/j.mhpa.2020.100379

Xie, Y., Gao, X., Song, Y., Zhu, X., Chen, M., Yang, L., & Ren, Y. (2021). Effectiveness of physical activity intervention on ADHD symptoms: A systematic review and meta-analysis. *Frontiers in Psychiatry*, *12*, Article 706625. https://doi.org/10.3389/fpsyt.2021.706625

Zarka, D., Cebolla, A. M., Cevallos, C., Palmero-Soler, E., Dan, B., & Cheron, G. (2021). Caudate and cerebellar involvement in altered P2 and P3 components of GO/NoGO evoked potentials in children with attention-deficit/hyperactivity disorder. *European Journal of Neuroscience*, *53*(10), 3447–3462. https://doi.org/10.1111/ejn.15198

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